

Training Data Reconstruction from Molecular Classifiers

If a model memorized its training data — can you make it confess?

AiChemist Team · Copenhagen Hackathon 2026

Membership Inference · ECFP4 · Neural Networks

The Challenge

Two black-box neural networks classify molecules by biological activity.

Goal: submit 1000 molecules per task that match the original training set.

Task 1 — Binary Classifier

- Architecture: 2048 → 512 → 128 → 2
- No dropout, no batch norm
- Train acc: ~100% Test acc: ~65%
- Deliberately overfit — memorized training data

Task 2 — 4-Class Classifier

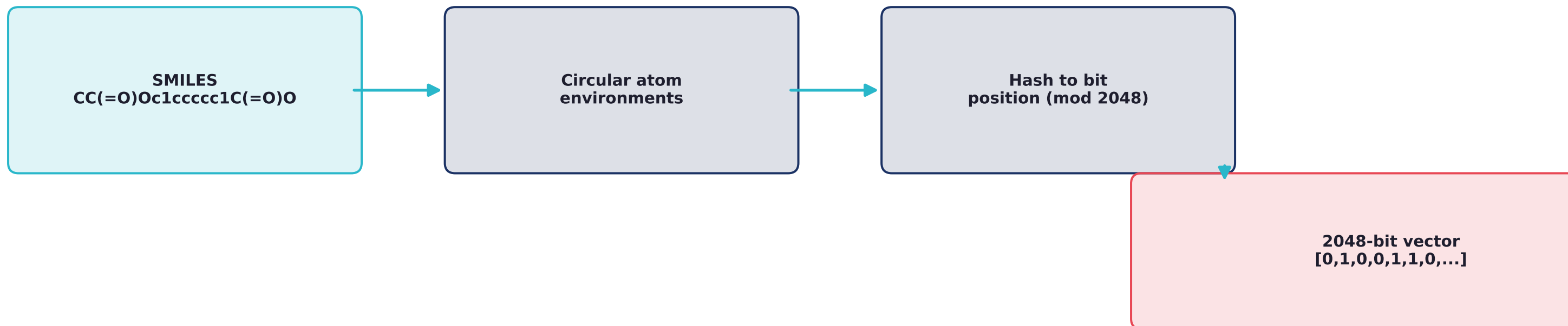
- Architecture: 2048 → [BN+ReLU+Dropout] → 512 → 256 → 4
- AdamW + early stopping (epoch 4)
- Train acc: ~90% Test acc: ~86%
- Regularized — generalizes well

Both models take ECFP4 fingerprints as input (Morgan, radius 2, 2048 bits)

ECFP4 Fingerprints

How molecules become numbers

Each molecule is encoded as a 2048-bit binary vector — its 'digital barcode'.

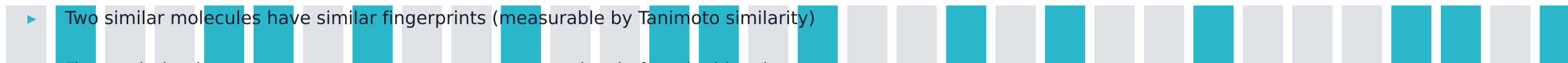


- ▶ Same molecule always produces the same fingerprint

- ▶ Two similar molecules have similar fingerprints (measurable by Tanimoto similarity)

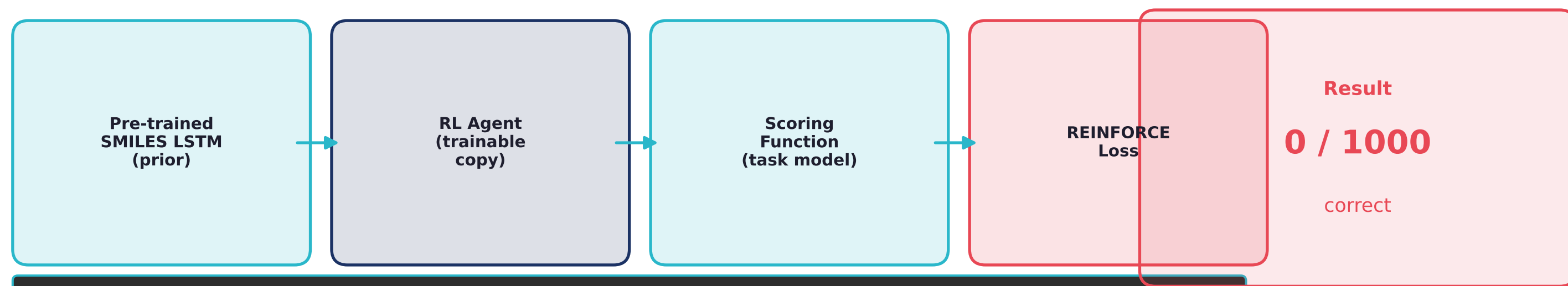
- ▶ Fingerprinting is one-way — you cannot reconstruct a molecule from its bits alone

← 32 of 2048 bits shown — each '1' means a specific chemical substructure is present



Approach 1 — Generative Pipeline

REINVENT RL (attempted first)



```
augmented_ll = prior_log_prob(smiles) +  $\sigma$  × confidence(smiles)
loss         = mean( ( augmented_ll - agent_log_prob(smiles) )2 )
→ agent learns to generate molecules the task model is confident about
```

The model generates valid, drug-like molecules — but not the specific training molecules.

A generative model produces plausible molecules from its learned distribution, not from the task model's memory.

The Key Insight — Raw Logits

Why softmax fails as a membership signal

Softmax Confidence

saturates — loses information

logit = 10	→	softmax = 0.9999	≈ 1.00
logit = 27	→	softmax = 0.9999	≈ 1.00
logit = 66	→	softmax = 1.0000	≈ 1.00

All look identical!

You cannot tell training members from outsiders.

Raw Logit

keeps full dynamic range

Wrong dataset
max logit ≈ 10



Closer dataset
max logit ≈ 27



Training source
max logit ≈ 66

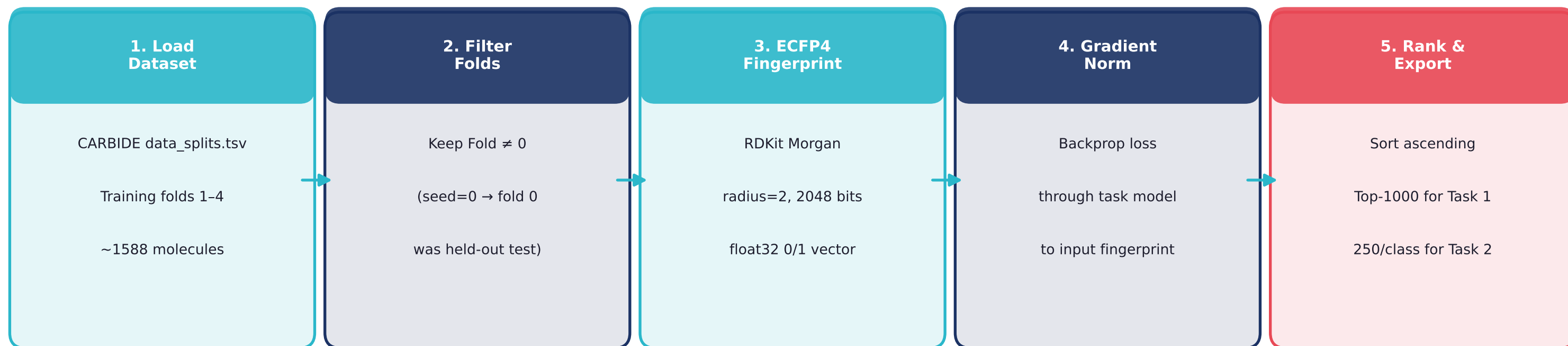


6x logit gap reveals training source

The model 'remembers' — and logits betray it.

Membership Inference Pipeline

The winning approach



```
x.requires_grad_(True)
loss = cross_entropy( model(x), predicted_class )
loss.backward()
gradient_norm = x.grad.norm()    # small norm  $\rightarrow$  loss minimum  $\rightarrow$  memorized
```

Why Gradient Norm Works

Mathematical intuition

The Mathematics

- ▶ Cross-entropy loss is minimized at training data points
- ▶ At a minimum, the gradient = 0 (definition of minimum)
- ▶ Non-training molecules are not at any minimum $\rightarrow$ gradient > 0

$$\begin{aligned}\partial \text{Loss} / \partial x &= (\text{softmax}(\text{logits}) - \text{one_hot}(y)) \cdot \partial \text{logits} / \partial x \\ &\approx 0 \quad \text{for training molecules} \quad (\text{loss} \approx 0) \\ &> 0 \quad \text{for non-training molecules}\end{aligned}$$

Therefore:

small gradient norm $\rightarrow$ loss minimum $\rightarrow$ memorized (training)

large gradient norm $\rightarrow$ not a minimum $\rightarrow$ unknown molecule

Loss Landscape (schematic)



Results & Journey

Random
Generation



0 correct

Prior generates valid
molecules — not training ones

Large
Database
Screening



0 correct

Wrong dataset — logits
all similar (~10)

Logit Signal
Discovery



60+ correct

Right dataset found via
logit magnitude jump

Gradient
Norm



Best result

Principled membership
inference on training folds

Key lesson

Raw logit magnitude (not softmax confidence) is the discriminative membership signal in overfit models. A 6× logit gap between the wrong and right dataset is unmistakable.

Takeaways

- 1 Dataset identification matters more than algorithm sophistication
- 2 Use raw logits — not softmax — for membership inference in overfit models
- 3 Gradient norm is the principled signal: training points sit at loss minima
- 4 A generative model cannot reconstruct specific training molecules
- 5 Fold structure in the dataset reveals the train/test split